



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

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Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

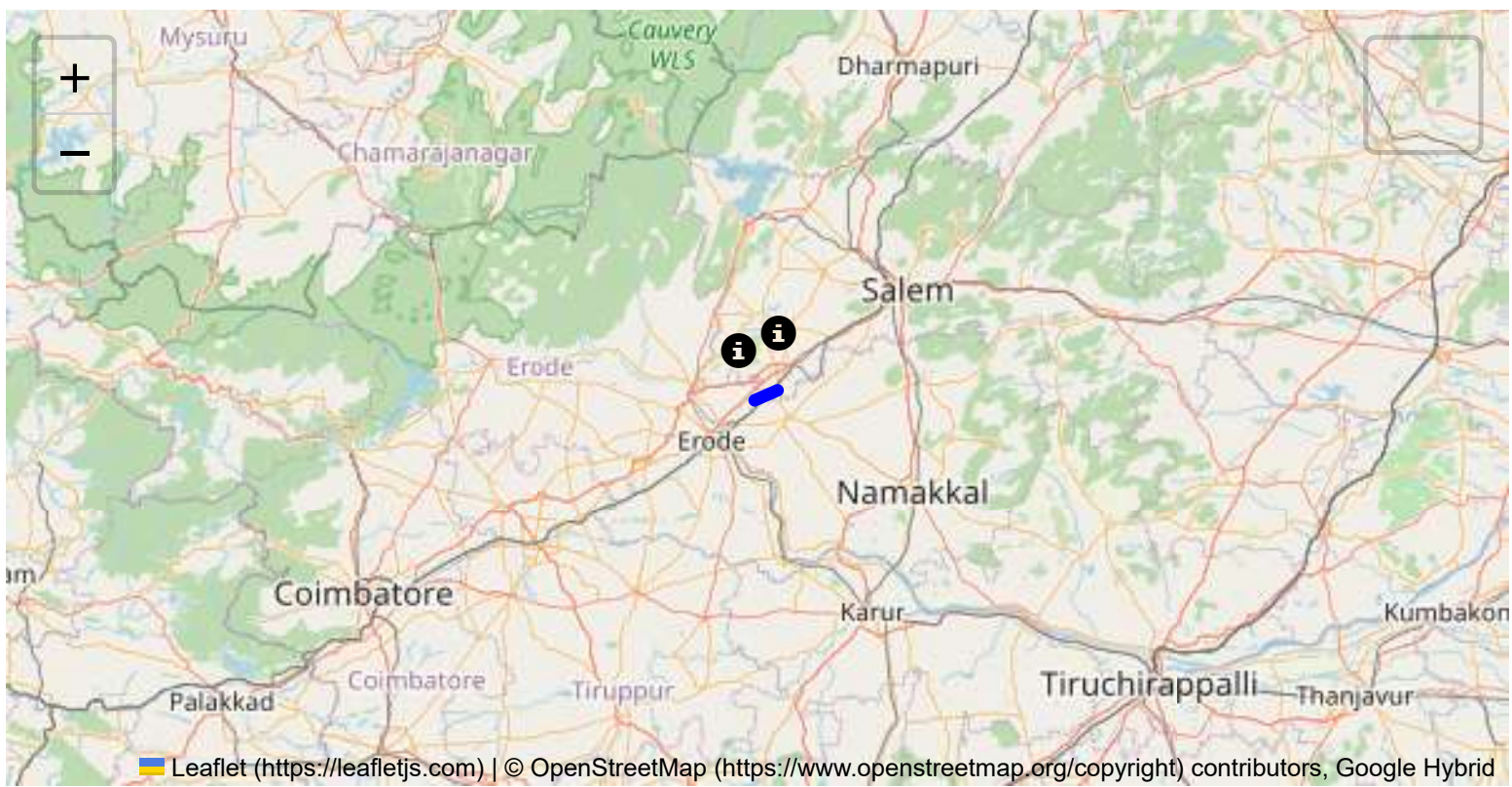
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

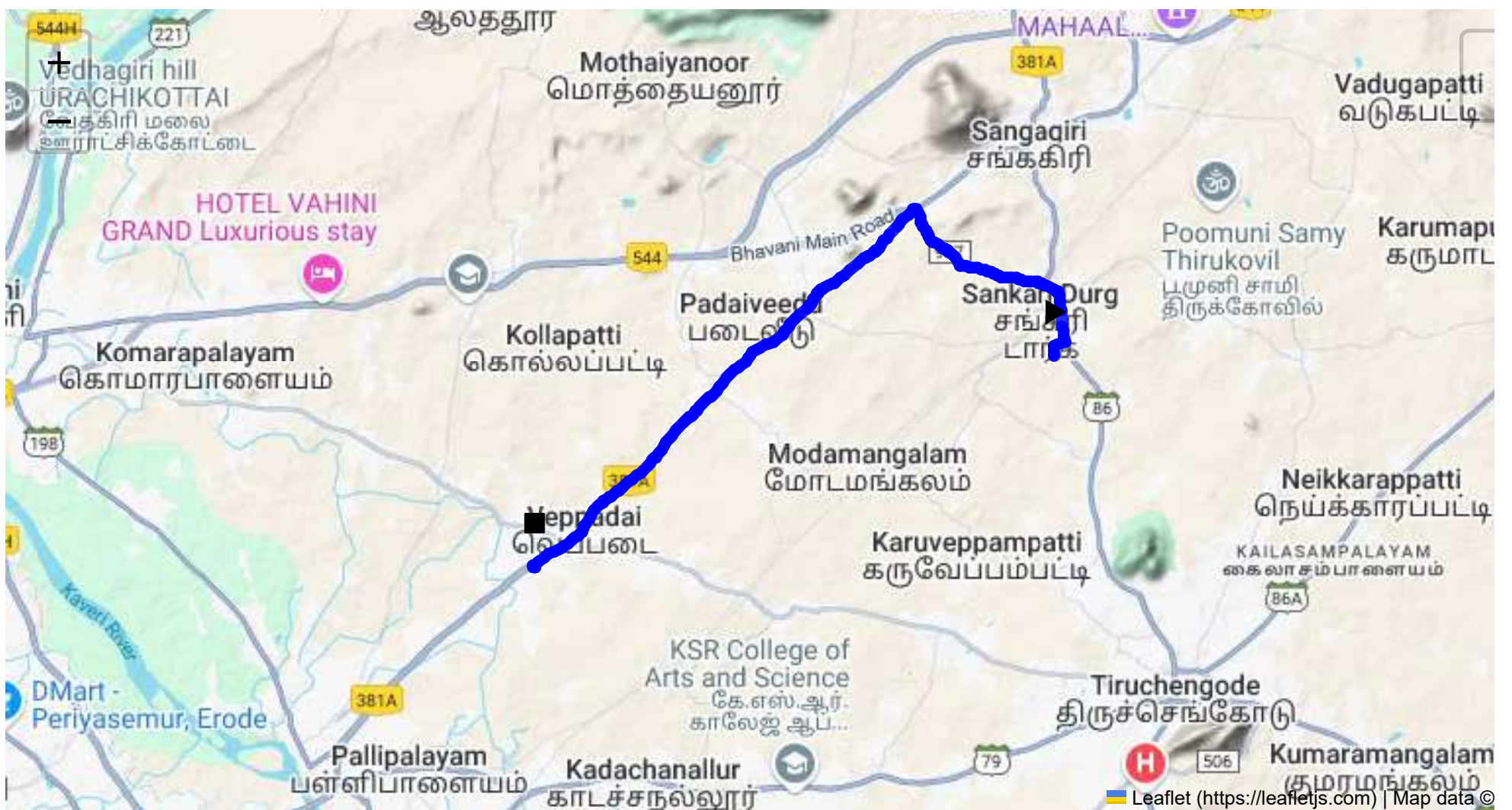
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 14.95 km
Estimated Duration: 0.3 hours
Adjusted Duration (Heavy Vehicle): 0.4 hours
Start: (11.4381, 77.8734)
End: (11.4016998, 77.78231924)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map:

The route between CVQF+23W, Sangagiri, Tamil Nadu, and CQ2J+PWX, Tiruchengode, Tamil Nadu, is approximately 14.95 kilometers, primarily along state highways and local roads. The path navigates through mostly rural areas with some small towns and agricultural land.

2. **Typical Weather Conditions and Potential Weather-Related Hazards:**

Tamil Nadu typically experiences moderate to heavy rainfall from October to December during the northeast monsoon. There is potential for waterlogging and slippery road conditions during this time. Summer months (March to June) can be extremely hot, posing risks of overheating, especially for vehicles transporting hazardous materials.

3. **Analysis of Traffic Patterns:**

Peak traffic hours in these areas are usually around 8:00-10:00 AM and 5:00-7:00 PM. Congestion is not as severe as in metropolitan areas but can occur near market areas and schools. Watch for slow-moving tractors and other agricultural vehicles.

4. **Assessment of Road Quality and Infrastructure:**

The primary highways are paved but may have irregularities like potholes, especially after the monsoon season. Rural roads may have less consistent maintenance, with sporadic bumps and narrow stretches that could pose challenges for heavy vehicles.

5. **Suggestions for Alternative Routes for Emergencies:**

In case of emergencies, an alternative route could involve taking the local roads towards Erode and joining larger highways from there. However, this could add distance and time to the journey.

6. **Summary of Local Regulations Affecting Hazardous Material Transport:**

The transportation of hazardous materials in Tamil Nadu is governed by national guidelines that mandate specific route permissions, necessary documentation, and adherence to speed limits. Night transport restrictions might apply.

7. **Overview of Historical Incidents:**

There is limited data on specific incidents due to the rural nature of the route, but in general, the region has seen accidents mostly related to overloaded vehicles and poor visibility during weather events.

8. **Environmental Considerations and Sensitive Areas:**

There are minor forest areas and agricultural fields along the route. Care should be taken to avoid fuel or material spills that could contaminate these areas. Wildlife is generally not dense but watch for stray cattle.

9. **Analysis of Communication Coverage:**

Mobile network coverage is generally reliable along state highways but may be spotty on more rural roads. Certain stretches through less populated areas might experience weak signals.

10. **Estimated Emergency Response Times:**

Emergency services in rural Tamil Nadu might be slower compared to urban centers. Response times can range from 15 to 30 minutes depending on proximity to the nearest town with services like hospitals or fire stations.

11. **Overall Summary of Risk Assessment:**

The route poses moderate risks mainly from weather-related road conditions and traffic patterns in small towns. Prepared readiness for sudden congestion, appropriate speed management, and contingency planning for adverse weather conditions are advised. Infrastructure is relatively sufficient, though vigilance is needed on rural roads and in dead zones. Familiarity with local

regulations regarding hazardous materials is crucial. Regular updates on route and weather conditions, along with emergency contact accessibility, will enhance route safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	Medium	11.43971, 77.87348	30 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.46307, 77.84941	30 KM/Hr
5	Turn	Medium	11.46318, 77.84913	30 KM/Hr
6	Turn	Medium	11.46312, 77.84886	30 KM/Hr
7	Turn	High	11.40198, 77.78225	15 KM/Hr
8	Turn	High	11.40172, 77.78236	15 KM/Hr

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46307, 77.84941



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46318, 77.84913



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46312, 77.84886



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.40198, 77.78225



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.40172, 77.78236

